The briefing: evidence, inference, prediction

LEARNING TARGET · R1 I can separate evidence from inference and name the measurements needed to predict a future position.

A · KEY VOCABULARY

Evidence	What was actually stated or measured. "An object is approaching."
Inference	A conclusion that still needs testing. "It will hit Earth."
Model	A simplified description with stated assumptions that makes a testable prediction.
Testable question	"Is it bad?" → "Will its predicted path intersect Earth's position at the same time?"

B · MODEL + WHAT A PREDICTION NEEDS

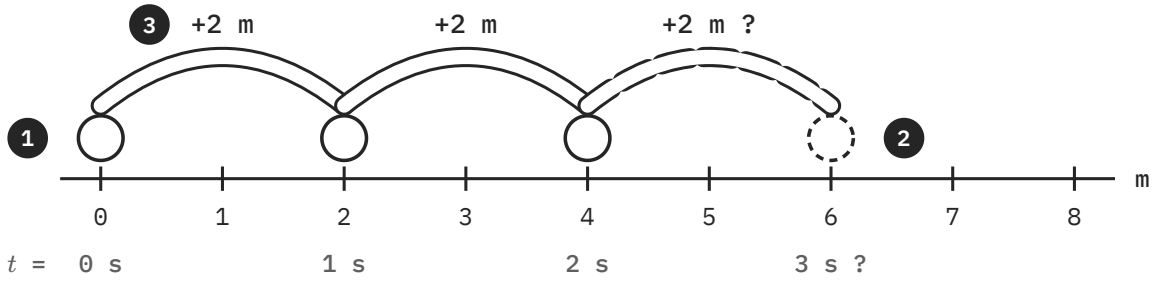
$$\text{new position} = \text{starting position} + \text{change} \times \text{elapsed time}$$

needs
- current position
- velocity *with direction*
- elapsed time + does v change?

CLAIM · EVIDENCE · REASONING · LIMIT

C · COLOR IT · WATCH A PATTERN FORM

One dot marks the cart's position each second.



- ☐ **1 · BLUE** Observations: measured positions
- ☐ **2 · BLUE** Prediction: dashed outline
- ☐ **3 · PURPLE** The pattern: +2 m each second

D · WORKED EXAMPLE

A cart is at 10 m, 8 m, and 6 m at 0, 1, and 2 s. Predict its position at 4 s, and name an observation that would challenge the model.

GIVEN	EQUATION	WORK	ANSWER
x = 10, 8, 6 m t = 0, 1, 2 s	new x = start + change × time	change = −2 m each second $x(4\text{ s}) = 6 + (-2)(2)$	**2 m** limit: a 3 m reading would challenge it

E · THE 2026-XJ THREAD

DAY 1 OF 22 · NEW TOOL: EVIDENCE VS. INFERENCE

The desk holds evidence (what was measured) and inference (what it suggests). A forecast for 2026-XJ must rest on the evidence: positions, times, and their uncertainty.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else